SOFTWARE REQUIREMENTS SPECIFICATION

Interest-Based Activity Discovery, Matching &
Group Coordination Platform

Version 1.0 | 12 October 2026

Prepared for product planning, design, engineering, and QA

One platform to help people discover activities, find compatible participants, organize plans, and join groups for everyday life.

Document Control

Field	Value
Document	Software Requirements Specification (SRS)
Version	1.0
Status	Initial baseline for review
Product category	Social activity discovery, matching, and coordination
Primary client	Mobile application; Android first, iOS-ready architecture
Proposed client technology	Flutter / Dart
Audience	Product owner, UI/UX designer, developers, QA, operations, and security reviewers

Revision History

Version	Date	Description
1.0	12 October 2026	Initial requirements baseline based on the product concept.

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for a mobile platform that helps people find others for activities they want to do. Users can state their interests, discover relevant activities, create a plan, specify time and place, invite or request participants, communicate with accepted participants, and manage the activity through completion.

1.2 Product Vision

Make it easy for a person to turn “I want to do this” into “I found the right people and we made a plan.” The platform supports solo-to-group discovery, filling open spots in an existing group, creating new activities, and joining public or private activities.

1.3 Intended Audience

- Product owner and stakeholders defining scope and priorities.
- UI/UX designers creating user flows, wireframes, and prototypes.
- Flutter developers building the mobile application.
- Backend developers implementing APIs, matching, chat, notifications, and data storage.
- QA engineers verifying requirements and acceptance criteria.
- Moderation, customer support, and operations teams managing safety and reports.

1.4 Definitions

Term	Definition
Activity	A planned event or action, such as cricket, tea, shopping, a trip, or a study session.
Host	A user who creates and manages an activity.
Participant	A user who requests to join or is accepted into an activity.
Domain/category	A user interest area, such as sports, food, travel, games, hobbies, or dating/social.
Matching	Filtering and ranking activities or users by interests, time, distance, eligibility, and preferences.
Open spots	The number of participants still permitted for an activity.
Poll	A question with selectable options used to coordinate group decisions.
Private activity	An activity visible only to invited users or approved members.
MVP	Minimum Viable Product: the smallest safe and useful initial release.

2. Overall Description

2.1 Product Perspective

The product is a location-aware social coordination platform, not merely a feed or chat application. Its core workflow is activity-first: users discover or create an activity, matching people are notified or can find it in discovery, interested users request to join, the host approves or the system admits them according to the activity's join policy, and participants coordinate safely.

2.2 User Classes

User class	Description	Main permissions
Guest	Not authenticated	View introductory screens and sign in/register; no posting or messaging.
Registered user	Person seeking or organizing activities	Manage profile/interests, discover activities, request/join, create activities, chat where permitted, report/block.
Host	Registered user who created an activity	Edit/cancel activity, review requests, manage participants, communicate with participants, mark completion.
Moderator/support	Authorized operational staff	Review reports, investigate abuse, remove content, restrict accounts, respond to support requests.
Administrator	Trusted platform operator	Manage categories, policies, roles, configuration, audit logs, and platform metrics.

2.3 Supported Activity Categories

- Sports and fitness: cricket, football, badminton, gym, running, cycling.
- Food and casual outings: tea/coffee, restaurants, street food, brunch.
- Travel and outdoors: trips, trekking, camping, sightseeing, road trips.
- Entertainment: movies, concerts, events, bowling, amusement parks.
- Shopping and errands: mall visits, fashion shopping, markets, shared errands.
- Games: board games, multiplayer video games, LAN sessions, esports.
- Learning and careers: study groups, projects, hackathons, workshops, networking.
- Creative hobbies: photography, dance, music, filmmaking, art, book clubs.
- Social and relationships: making friends, group dates, double dates, singles' events.
- Family, community, volunteering, pets, and other user-suggested categories.

Categories must be configurable from the backend. Users may suggest a new category, but platform administrators or a controlled taxonomy process should approve it to prevent duplicates and unsafe categories.

2.4 Operating Environment

- Mobile application built with Flutter/Dart, initially targeting Android and designed to support iOS.
- Backend exposed through authenticated HTTPS APIs.
- Database for accounts, interests, activities, memberships, messages, reports, and notification preferences.
- Push notifications for requests, approvals, activity changes, chat, reminders, and safety alerts.
- Optional map/geocoding provider for selecting a public meeting point and estimating distance.
- Administrative web console for moderation and category/configuration management.

2.5 Assumptions and Constraints

- A network connection is required for sign-in, discovery, joining, messaging, and synchronization.
- Location access is optional; users can manually select an area or enter a venue.
- Exact live location is not shared with other users by default.
- The platform must define minimum age, dating eligibility, identity verification levels, and region-specific legal obligations before launch.
- Activity availability depends on users in the area; low-density locations may have fewer matches.
- Third-party services for maps, SMS/email, push delivery, and identity verification may impose cost and usage limits.

3. Scope and Release Strategy

3.1 MVP Scope

- Registration, login, logout, password reset, basic profile, and account deletion request.
- Interest onboarding with selectable categories and subcategories; users can edit preferences later.
- Home feed and discovery filters by category, date/time, approximate distance, group size, and open spots.
- Create, view, edit, cancel, and report an activity.
- Activity details including purpose, time, meeting place/venue, capacity, estimated cost, rules, and join policy.
- Request to join, approve/reject requests, leave activity, and prevent over-capacity.
- Activity participant list and activity-scoped group chat after joining/approval.
- Push/in-app notifications for requests, decisions, cancellations, changes, and reminders.
- Block/report users and activities; moderator review tools.
- Basic admin controls for categories, content reports, and account restrictions.

3.2 Post-MVP Candidates

- Personalized recommendation ranking and compatibility scoring.
- Polls for selecting times, venues, budgets, and activity options.
- Recurring activities, clubs, communities, teams, and public event series.
- Calendar integration, waitlists, QR check-in, attendance, and post-activity reviews.
- Optional verification badges, enhanced trust signals, and fraud prevention.
- Payments, ticketing, deposits, refunds, and paid event support only after legal/security review.
- Web client, advanced analytics, multilingual support, and AI-assisted recommendations.

3.3 Out of Scope for Initial MVP

- Guaranteed identity or background verification for every user.

- Emergency-response services or a guarantee that an in-person meeting is safe.
- Real-money betting or gambling.
- Automatic disclosure of a user's exact live location to strangers.
- Unmoderated anonymous direct messaging to arbitrary users.
- A promise that every activity will find enough participants.

4. User Journey and Core Workflows

4.1 First-Time User

1. User opens the app and sees introductory/onboarding screens.
2. User registers or signs in and accepts the terms, privacy notice, and community/safety rules.
3. User creates a profile with display name, profile image (optional), age eligibility data as required, and a short bio.
4. User selects interests and sub-interests, for example Sports → Cricket; Food → Tea/Coffee; Travel → Weekend trips.
5. User sets discovery preferences such as preferred activity radius, notification preferences, and optional availability.
6. App opens the personalized Home/Discover screen; the user can change interests and privacy settings at any time.

4.2 Example: Organize a Cricket Match

1. Host selects Create Activity → Sports → Cricket.
2. Host enters date, start/end time, venue or meeting point, player count, skill level (optional), cost per person (optional), and rules.
3. Host chooses visibility (public or invite-only), join policy (approval required or auto-join), and target audience/preferences.
4. System validates fields and publishes the activity.
5. Eligible users who follow cricket and meet the discovery criteria can see the activity; matching notifications are sent only according to user preferences and rate limits.
6. Interested users open details and tap Request to Join or Join.
7. Host reviews requests when approval is required. System updates available spots atomically to prevent overbooking.
8. Accepted participants receive confirmation and can use the activity group chat.
9. System sends configurable reminders and notifies participants of schedule or venue changes.
10. Host marks the activity completed or cancelled; participants may optionally provide feedback or report a problem.

4.3 Example: Invite People for Tea/Coffee

1. User selects Create Activity → Food & Drink → Tea/Coffee.
2. User chooses cafe or public meeting point, date/time, participant limit, estimated spend, and whether the plan is one-to-one or a small group.
3. The activity is shown to eligible users who opted into relevant social/food activities and match the visibility and preference rules.
4. Interested users request to join; the host can review profile basics and accept or decline.
5. Contact details remain private unless a user deliberately shares them; communication occurs through the app after the relevant join/approval condition.
6. Both sides can cancel, block, or report. Safety reminders encourage meeting in public and telling a trusted person where appropriate.

4.4 General Activity Lifecycle

State	Meaning	Allowed transitions
Draft	Created but not published	Published or Deleted
Published / Open	Visible and accepting participants	Full, In progress, Cancelled, Expired
Full	Capacity reached; new requests may enter waitlist if enabled	Open if a spot opens, In progress, Cancelled
In progress	Activity start time has arrived or host starts it	Completed, Cancelled
Completed	Activity finished	Closed; feedback/reporting may remain available
Cancelled	Host or admin cancelled activity	Terminal except audit/history
Expired	Activity time passed without completion action	Closed or host-confirmed completion, according to policy

5. Functional Requirements

Priority legend: P0 = required for MVP; P1 = important next release; P2 = future enhancement. Each requirement has a stable identifier for development and QA traceability.

ID	Area	Priority	Requirement
FR-001	Authentication	P0	The system shall allow a user to register using an approved authentication method (email/phone or supported identity provider).
FR-002	Authentication	P0	The system shall authenticate users and provide logout, session expiry, and password/account recovery where applicable.
FR-003	Authentication	P0	The system shall require acceptance of current Terms of Service, Privacy Notice, and Community Guidelines before participation features are enabled.
FR-004	Profile	P0	The system shall allow users to create and edit a profile with display name, bio, profile image, interests, and relevant safety/privacy settings.
FR-005	Profile	P0	The system shall let users deactivate or request deletion of their account and explain the data-retention/deletion process.
FR-006	Interest onboarding	P0	The system shall present a searchable hierarchy of activity categories and subcategories during onboarding.
FR-007	Interest onboarding	P0	The system shall let users select multiple interests, remove interests, and update interests after onboarding.
FR-008	Discovery	P0	The system shall display a feed of relevant activities and support search by keyword/category.
FR-009	Discovery	P0	The system shall support filtering by date/time, approximate location or area, distance/radius when available, open spots, activity type, and budget when supplied.
FR-010	Discovery	P1	The system should rank activities using selected interests, proximity, time fit, capacity, freshness, and user preference while avoiding discriminatory or opaque sensitive-attribute inference.
FR-011	Activity creation	P0	The system shall allow an authenticated user to create an activity with title, category, description, date/time, location/venue, minimum/maximum participants, visibility, join policy, and host.
FR-012	Activity creation	P0	The system shall support optional fields such as cost estimate, skill level, age eligibility, equipment needed, rules, cover image, and accessibility notes.
FR-013	Activity creation	P0	The system shall validate required fields, date/time consistency, participant limits, category, and location before publication.
FR-014	Activity management	P0	The host shall be able to edit or cancel an activity and the system shall notify affected participants of material changes.
FR-015	Activity management	P0	The system shall show activity status, capacity, accepted participants, pending requests, and open spots to authorized users.
FR-016	Matching/targeting	P0	The system shall make published activities discoverable to users matching category/interests and the activity's visibility and eligibility rules.
FR-017	Matching/targeting	P0	The system shall respect user notification preferences and shall not send an unsolicited message to every user merely because they share a broad interest.

ID	Area	Priority	Requirement
FR-018	Join flow	P0	A user shall be able to join immediately when auto-join is enabled and capacity/eligibility checks pass, or request to join when host approval is required.
FR-019	Join flow	P0	The host shall be able to approve or reject pending requests and the requester shall receive the decision.
FR-020	Join flow	P0	The system shall prevent a participant from occupying duplicate slots and shall prevent accepted participants from exceeding the maximum capacity.
FR-021	Join flow	P1	The system should support a waitlist and offer a newly available spot to the next eligible user when enabled.
FR-022	Join flow	P0	A participant shall be able to withdraw a pending request or leave an activity subject to clearly displayed timing/cancellation rules.
FR-023	Polls	P1	The system should allow hosts or participants to create polls for options such as time, venue, budget, or activity choice, with defined closing time and voting rules.
FR-024	Messaging	P0	The system shall provide activity-scoped group chat to approved participants, subject to activity state and moderation rules.
FR-025	Messaging	P1	The system may support one-to-one messaging only where the recipient has opted in, a request has been accepted, or another explicit permission rule is met.
FR-026	Notifications	P0	The system shall provide in-app notifications for join requests, decisions, activity changes, cancellations, chat messages, and relevant reminders.
FR-027	Notifications	P0	Users shall be able to control non-essential notification categories and frequency.
FR-028	Location	P0	The system shall allow manual venue/area selection if location permission is denied.
FR-029	Location	P0	The system shall display a public venue or approximate area to prospective participants and shall not expose a user's precise live location by default.
FR-030	Trust & safety	P0	Users shall be able to block another user and report a user, message, activity, or profile.
FR-031	Trust & safety	P0	The system shall hide or restrict blocked-user interactions according to policy and prevent the blocked user from contacting the blocker through supported app channels.
FR-032	Moderation	P0	Authorized moderators shall be able to review reports, remove content, warn/restrict/suspend accounts, and record moderation reasons.
FR-033	Moderation	P0	The system shall retain an auditable record of material moderation and administrative actions, with access controls.
FR-034	Reviews	P1	After an activity ends, the system should allow participants to submit structured feedback and report safety concerns; public ratings must be protected against abuse.
FR-035	Admin	P0	Administrators shall be able to manage categories, subcategories, activity policies, and account roles.
FR-036	Admin	P0	Administrative access shall require role-based authorization and stronger authentication appropriate to risk.
FR-037	Privacy	P0	Users shall be able to review and change profile visibility, discoverability, and optional location-sharing settings.
FR-038	Privacy	P0	The system shall collect only data necessary for defined product functions and document retention, export, and deletion policies.
FR-039	Reliability	P0	The system shall preserve the final participant count under concurrent join requests and return a clear result when an activity becomes full.
FR-040	Accessibility	P0	Core flows shall support screen readers, scalable text, adequate contrast, and touch targets suitable for mobile use.
FR-041	Account safety	P0	The system shall rate-limit sign-up, login attempts, posting, joining requests, invitations, and messaging to reduce spam and abuse.
FR-042	Activity reporting	P0	The system shall allow a user to report an activity as misleading, unsafe, inappropriate, spam, or otherwise policy-violating.
FR-043	Audit/history	P1	The system should retain a user's activity history and participation status according to privacy and retention settings.
FR-044	Calendar	P2	The system may allow users to add confirmed activities to a device calendar with explicit permission.
FR-045	Payments	P2	If payments are later introduced, the system shall define payment status, provider integration, refunds, disputes, receipts, and applicable compliance before release.

6. Business Rules

- Every activity must have one host, a category, a start time, a location/meeting method, and a maximum participant count.
- Minimum participants must be less than or equal to maximum participants. The minimum is an activity goal; the maximum is a hard capacity limit.
- The host may optionally require a minimum number of participants before the activity is considered confirmed. The app must show the confirmation status clearly.
- An activity can be public/discoverable, private/invite-only, or unlisted/share-link based. Visibility and invitation permissions must be explicit.
- The platform must distinguish a user's interest from consent to receive direct messages or repeated invitations.
- Interest membership must not automatically expose phone number, email, exact location, or other private profile data.
- Users must be able to report or block without needing to confront the other user.
- For dating or romantic categories, participation must follow the product's age policy, consent rules, anti-harassment policy, and applicable law. Dating must not be silently enabled for ineligible users.
- The platform must define cancellation behavior, no-show handling, waitlist behavior, and what happens when a host account is suspended.
- Repeated invitations must be rate-limited; users must not be able to broadcast unlimited requests to everyone in a domain.
- The system must prevent unauthorized users from viewing private activities, private chats, or pending requests.
- Activity recommendations must explain key relevance signals where practical and must not infer sensitive traits from unrelated activity.

7. Data Requirements and Conceptual Data Model

Entity	Key fields (illustrative)	Purpose
User	user_id, auth_subject, display_name, bio, avatar_url, age/eligibility attributes where required, status, created_at	Account and public profile
Interest	interest_id, parent_id, name, status	Configurable category hierarchy
UserInterest	user_id, interest_id, preference_strength, created_at	Many-to-many user interests
Activity	activity_id, host_id, title, description, category_id, start_at, end_at, venue, approximate_location, min_participants, max_participants, visibility, join_policy, status	Activity listing and lifecycle
ActivityMembership	membership_id, activity_id, user_id, role, status, requested_at, decided_at	Requests, accepted members, host, waitlist
Poll	poll_id, activity_id, creator_id, question, closes_at, status	Group decisions
PollOption / Vote	option_id, poll_id, label; vote_id, option_id, user_id	Poll choices and votes
Conversation / Message	conversation_id, activity_id, message_id, sender_id, body, created_at, moderation_status	Activity-scoped communication
Notification	notification_id, user_id, type, payload_reference, read_at, created_at	In-app and push notification state
Block	blocker_id, blocked_id, created_at	User safety preferences
Report	report_id, reporter_id, target_type, target_id, reason, details, status, created_at	Safety/moderation workflow
Review	review_id, activity_id, reviewer_id, target_id, rating/structured_feedback, created_at	Post-activity feedback
AdminAuditLog	log_id, actor_id, action, target_type, target_id, reason, created_at	Traceability for privileged actions

This is a conceptual model, not a final database schema. The implementation team should define indexes, uniqueness constraints, deletion behavior, data retention, and transaction boundaries during detailed design.

7.1 Important Data Constraints

- Unique membership constraint for (activity_id, user_id) to prevent duplicate membership.
- Capacity updates must use a transaction, lock, or equivalent atomic conditional update.
- All times must be stored consistently (prefer UTC) and rendered in the user's local timezone.
- Location should be stored at the minimum precision necessary; exact home addresses should not be required for ordinary activities.
- Private message and report data must have strict authorization checks and retention rules.
- Deletion/anonymization must account for legal obligations, abuse prevention, and records that cannot lawfully be erased immediately.

8. External Interface Requirements

8.1 User Interface

- Authentication: welcome, register, sign in, recovery, consent.
- Onboarding: profile basics, category selection, location preference, notification permissions.
- Home/Discover: recommended activities, categories, search, filters, create button.
- Activity details: host, schedule, public meeting point, participant count, rules, join/request action, report action.
- Create/edit activity: category-driven fields, capacity, visibility, join policy, venue picker, preview.
- Requests/participants: pending, accepted, rejected, waitlisted, leave/cancel controls.
- Activity chat and polls: membership-gated, report/block affordances, clear system notices.
- Notifications, profile/interests, settings/privacy, blocked users, help/safety center.
- Moderator/admin console: reports queue, report details, actions, category management, audit history.

8.2 API and Integrations

- Versioned HTTPS REST API (or equivalent documented API) for account, profile, interest, activity, membership, poll, message, notification, and moderation resources.
- Push notification provider and optional email/SMS provider.
- Maps/geocoding provider for venue selection; avoid sending unnecessary personal data to third parties.
- Object storage/CDN for profile and activity images, with file-type/size validation and abuse scanning.
- Optional analytics/crash reporting with privacy-aware event design and no secrets or sensitive message content in logs.

8.3 Error Handling

The app must show actionable, human-readable errors for network failure, expired session, permission denial, full activity, closed activity, rejected request, invalid time, unavailable venue service, upload failure, and rate limiting. Retry actions must not duplicate activities, memberships, messages, or payments.

9. Non-Functional Requirements

ID	Quality attribute	Priority	Requirement
NFR-001	Performance	P0	For typical supported network conditions, common API reads should target p95 under 500 ms excluding third-party latency; activity creation/join actions should target p95 under 800 ms. Validate with load tests.
NFR-002	Responsiveness	P0	Primary screens should render useful content promptly and provide loading, empty, offline, and error states.
NFR-003	Availability	P1	Production backend target should be defined with the hosting/SLO plan; initial target recommendation is 99.5% monthly availability, excluding planned maintenance.
NFR-004	Scalability	P0	Architecture should support horizontal scaling of stateless API services and indexed/paginated discovery queries.
NFR-005	Security	P0	Use TLS in transit, secure password/auth-provider practices, short-lived access tokens where applicable, protected secrets, server-side authorization, and secure storage on device.
NFR-006	Privacy	P0	Use data minimization, purpose limitation, user-controlled visibility, retention/deletion policy, and appropriate consent for permissions and communications.
NFR-007	Authorization	P0	Enforce permissions on the server for every protected resource; hiding UI controls alone is insufficient.
NFR-008	Concurrency	P0	Concurrent join requests must not exceed activity capacity or create duplicate membership.
NFR-009	Reliability	P0	Important actions should be idempotent or protected against duplicate submissions caused by retries.
NFR-010	Accessibility	P0	Support screen readers, text scaling, sufficient contrast, focus order, and accessible labels for essential controls.
NFR-011	Maintainability	P0	Use modular feature-based client architecture, documented API contracts, automated tests, linting, and code review.
NFR-012	Observability	P0	Capture structured operational logs, metrics, crash/error signals, and correlation IDs while excluding passwords, tokens, and private message contents.
NFR-013	Localization	P1	Keep user-facing strings externalized and support locale-aware dates, times, and currencies.
NFR-014	Compatibility	P0	Define and test supported Android versions/devices before release; keep iOS support feasible through Flutter.
NFR-015	Backup & recovery	P1	Define backup frequency, restore tests, recovery objectives, and incident response procedures.
NFR-016	Abuse resistance	P0	Use rate limits, spam detection, report workflows, content policies, and escalation paths appropriate to the risk.
NFR-017	Data integrity	P0	Validate all client inputs on the server and enforce database constraints for critical invariants.
NFR-018	Testability	P0	Business rules must be testable through unit, integration, API, and end-to-end tests.

10. Security, Privacy, and Trust & Safety

10.1 Security Controls

- Use a reputable authentication solution or correctly implemented server-side authentication; never store plaintext passwords.
- Apply least-privilege role-based access for users, hosts, moderators, and administrators.
- Validate and authorize every activity, membership, chat, report, and admin request on the server.
- Protect against injection, broken access control, credential stuffing, spam, malicious uploads, and replay/duplicate requests.
- Use rate limits, abuse monitoring, secure session revocation, and audit logs for privileged actions.
- Do not expose secrets in the Flutter app, source control, crash reports, or client-side environment files.

10.2 Safety by Design

- Encourage public meeting venues for first-time in-person meetings and show practical safety guidance.
- Do not reveal exact live location, private phone number, or home address by default.
- Provide visible block/report controls on profiles, activities, and chat.
- Give moderators clear categories, severity levels, evidence access rules, and escalation procedures.
- Support cancellation and leaving an activity without forcing a user to explain themselves to other participants.
- Establish policies for harassment, sexual misconduct, threats, scams, minors, impersonation, discrimination, and dangerous activities.
- Before release, define a minimum-age policy and separate eligibility rules for dating/romantic categories; obtain legal/privacy review for launch jurisdictions.

10.3 India Launch Considerations

Because an initial launch may be in India, the product owner should obtain qualified legal advice on applicable privacy/data-protection requirements, intermediary/platform obligations where applicable, consumer protection, online safety, and age-related requirements. This SRS does not constitute legal advice. The privacy notice, grievance/contact process, retention policy, and user reporting process must be finalized before public launch.

11. Recommended Technical Architecture

The architecture should keep the Flutter client thin and place matching, authorization, capacity checks, and safety enforcement on the backend.

Layer	Responsibility	Implementation direction
Mobile client	Screens, navigation, forms, local state, cached non-sensitive data, permission prompts	Flutter/Dart; feature-first modules; state management selected by the team
API/backend	Authentication integration, business rules, activity lifecycle, matching, join transactions, chat authorization, moderation	REST API with versioning; modular monolith is a practical MVP starting point
Database	Users, interests, activities, membership, polls, messages, reports, audit	Relational database such as PostgreSQL; spatial extension/querying if required
Realtime	Activity chat and timely membership/status updates	WebSocket/realtime service or managed equivalent with server-side authorization
Notifications	Push and in-app notifications	FCM/APNs or managed provider; queue/retry and preference checks
Media	Profile and activity images	Object storage plus CDN, validation, access policies
Admin	Moderation and taxonomy management	Separate protected web console or tightly restricted admin interface
Operations	Monitoring, backups, secrets, deployment, analytics	CI/CD, environment separation, logs/metrics, backups, incident response

Avoid starting with many microservices. A well-modularized backend with clear boundaries is easier to ship, test, and evolve at MVP scale. Extract services only when operational evidence justifies it.

12. High-Level Screen Inventory

Screen	Purpose	MVP
Splash / Welcome	Brand, value proposition, login/register entry	Yes
Register / Login / Recovery	Secure account access	Yes
Profile setup	Create user profile and consent choices	Yes
Interest selection	Choose multiple domains and subdomains	Yes
Home / Discover	Recommended and nearby/relevant activities	Yes
Search / Filters	Find by keyword, category, time, distance, spots, budget	Yes
Activity details	Understand plan, host, venue, capacity, and rules	Yes
Create / Edit activity	Publish and manage an activity	Yes
Join requests / Participants	Approve, reject, leave, or view membership status	Yes
Activity chat	Coordinate with accepted participants	Yes
Notifications	See requests, decisions, updates, reminders	Yes
Profile / Interests / Privacy	Edit preferences and privacy controls	Yes
Report / Block / Help	Safety and moderation access	Yes
Polls	Coordinate choices within an activity	Later
Communities / Clubs	Persistent interest-based groups	Later
Reviews / Attendance	Post-activity feedback and check-in	Later
Admin moderation console	Review reports and manage taxonomy	Required before public launch

13. Acceptance Criteria for MVP

1. A new user can register, accept required policies, create a profile, select multiple interests, and reach the Discover screen.
2. A user can create and publish a cricket activity with time, venue, capacity, and join policy.
3. A user who follows cricket can discover the activity when its visibility, filters, and eligibility rules permit.
4. An interested user can request to join; the host can approve or reject; both receive the correct status and notification.
5. The system never admits more accepted participants than the activity's configured capacity, including during concurrent requests.
6. Approved participants can access the activity chat; non-members and unauthorized users cannot.
7. A host can edit or cancel an activity and accepted participants receive a clear update.
8. A user can create a tea/coffee outing and select a public cafe or meeting point without exposing their private home address.
9. Users can block/report profiles and activities; the action is recorded and available to authorized moderators.
10. Users can edit interests and notification preferences; non-essential broadcast notifications respect these settings.
11. Critical workflows have automated tests and pass manual testing on the defined supported Android devices.
12. The production deployment has monitoring, backups, rate limiting, secure configuration, a privacy notice, community guidelines, and an operational moderation process.

14. Testing Strategy

- Unit tests: validation, eligibility, matching filters, capacity logic, lifecycle transitions, and permission rules.

- API/integration tests: authentication, create/edit/cancel, request/approve/reject, membership authorization, notifications, reports, and deletion workflows.
- Concurrency tests: simultaneous join requests at the last available spot; duplicate retries; duplicate membership attempts.
- Security tests: IDOR/broken access control, unauthorized chat access, rate limits, token handling, upload validation, and admin privilege boundaries.
- UI tests: onboarding, interest selection, search/filter, activity creation, join flow, chat, settings, block/report, empty/error/offline states.
- Accessibility tests: screen reader labels, text scaling, contrast, keyboard/focus behavior where applicable.
- Performance tests: feed pagination, search latency, high-demand activities, notification fan-out, and chat load.
- Operational tests: deployment rollback, backup restore, alerting, moderation escalation, and account suspension.

15. Risks and Mitigations

Risk	Impact	Mitigation
Cold-start / low local user density	Users see few matching activities	Launch in a limited geography or community; seed legitimate events; allow invite links and clear availability indicators.
Spam and notification fatigue	Users disable notifications or leave	Opt-in notification preferences, per-user rate limits, relevance thresholds, mute controls, and digest options.
Harassment or unsafe meetings	User harm and reputational/legal risk	Safety-first defaults, reporting/blocking, moderation coverage, age rules, public meeting guidance, and incident escalation.
Fake profiles and scams	Trust erosion	Verification options, abuse signals, reporting, rate limits, and staged trust features.
Overbooking	Activity disruption	Atomic capacity enforcement, unique constraints, waitlists, and concurrency tests.
Too many categories / poor discovery	Confusing onboarding and irrelevant feed	Curated taxonomy, search, subcategories, editable interests, and measured recommendation quality.
Privacy/location leakage	Potential personal safety issue	Approximate/public venue display, explicit permission, data minimization, access controls, and privacy review.
Scope too broad for first release	Delayed launch and unstable product	Ship the defined MVP first; defer payments, advanced AI, clubs, and complex verification.

16. Suggested Implementation Phases

Phase	Deliverables	Exit condition
Phase 0 — Product definition	Finalize target users, age policy, initial launch geography, category taxonomy, safety policies, MVP metrics	Scope and key policies approved
Phase 1 — UX and architecture	User flows, wireframes, design system, API contracts, data model, threat model	Prototype and architecture reviewed
Phase 2 — Foundation	Flutter app structure, backend environment, authentication, profile, interests	User can register and complete onboarding
Phase 3 — Activity core	Create/edit/cancel, discovery, filters, details, join/approval, capacity control	End-to-end activity lifecycle passes tests
Phase 4 — Coordination	Activity chat, notifications, reminders, polls if included	Accepted members coordinate reliably
Phase 5 — Trust and admin	Block/report, moderation queue, admin roles, audit trail, privacy controls	Safety operations ready for closed beta
Phase 6 — Beta and launch	Device testing, security review, load testing, monitoring, policies, store release preparation	Release checklist approved
Phase 7 — Growth	Recommendation improvements, communities, recurring events, calendar, advanced trust features	Prioritized by real usage data

17. Product Success Metrics

- Activation: percentage of new users who complete onboarding and select at least one interest.
- Time to first value: time from registration to first activity view, join request, or created activity.
- Match success: percentage of activities that reach their minimum target before start time.
- Join conversion: activity detail views that become join requests or confirmed joins.
- Attendance/fulfillment: percentage of confirmed activities marked completed or attended.
- Retention: returning users at 7-day and 30-day intervals.
- Safety: report rate, time to first moderation action, repeat-abuse rate, and user block rate.
- Notification quality: opt-out rate, mute rate, and conversion from relevant notifications.
- Local liquidity: number of active activities and eligible participants by category and area.

Metrics should be aggregated with privacy safeguards. Do not optimize only for message volume or time spent; successful real-world activity completion and user safety are more meaningful outcomes.

18. Open Decisions Before Development

Decision	Recommended starting point	Owner / status
Product name and branding	Choose and validate a unique name before public launch	Product owner — Open
Initial geography	Start with one city/college/community or a small number of launch zones	Product owner — Open
Minimum age and dating policy	Define explicit age eligibility and dating-category safeguards before implementation	Product/legal — Must resolve
Authentication methods	Email/phone plus a trusted provider as appropriate for target market	Product/engineering — Open
Public vs invite-only defaults	Public discoverability with explicit visibility controls; private by choice	Product owner — Open
Join approval defaults	Host approval for social/meetup activities; optional auto-join for suitable low-risk activities	Product/safety — Open
Location precision	Public venue or approximate area; exact live location off by default	Product/safety — Recommended
Chat access	Activity-scoped chat for accepted members; direct messaging gated by consent/permission	Product/safety — Recommended
Minimum participant handling	Show target/minimum separately from hard maximum; decide auto-cancel/waitlist behavior	Product owner — Open
Backend hosting and database	Select based on team skills, budget, scale, compliance, and operational needs	Engineering — Open
Monetization	Defer until core activity completion and retention are validated	Product owner — Open
Moderation coverage	Define response times, escalation contacts, and severe-report handling before public launch	Operations — Must resolve

Appendix A — Example Activity Record

Field	Example
Title	Friendly Cricket Match — Saturday
Category	Sports > Cricket
Host	User 1042
Start / end	Saturday, 4:00 PM–6:00 PM local time
Venue	Selected public cricket ground
Minimum participants	10
Maximum participants	14
Accepted participants	8
Open spots	6
Join policy	Host approval
Visibility	Public to eligible users
Estimated cost	Optional; shown as estimate, not a payment
Status	Published / recruiting

Appendix B — Requirement Traceability

Every FR/NFR identifier should be linked in the project tracker to one or more design screens, API endpoints, database constraints, test cases, and release status. A requirement is not complete merely because its UI exists; server-side validation, authorization, error handling, and tests must also be delivered.